Preregistration Record: Block I B2 Causal Maintenance Network

Theory of Everything Research Programme

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Protocol ID: `B2_CAUSAL_MAINTENANCE_PREREG_001`
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Software baseline: `0.7.0 / STAGE_B2_0_7_0_PREREG_CANDIDATE`
Related experimental programme: 10.5281/zenodo.22857230

Registration status. This document was prepared for public OSF preregistration before any confirmatory B2 execution. At the time this record was frozen, all seven confirmatory manifests were marked `FROZEN_PREREGISTRATION_READY_UNRUN` and the confirmatory execution gate remained closed. The public OSF registration DOI will be assigned by OSF when the registration is made public and should be taken from the registration metadata.

1 Purpose and Scientific Scope

B2 is the first experiment in the Theory of Everything Research Programme that directly tests the causal distinction between persistence and internally generated maintenance. It asks whether an artificial system can preserve its operational organisation through identifiable internal maintenance processes rather than merely persisting because of slow degradation or continuous external support.

The candidate transition is therefore

$$\boxed{\text{persistent organisation} \longrightarrow ? \longrightarrow \text{internally maintained organisation.}} \tag{1}$$

A positive B2 result would support only artificial functional and causal self-maintenance within the registered architecture family. It would not establish biological life, metabolism in the biological sense, organisational closure as a universal necessity, the historical origin of life, heredity, intelligence, consciousness, or a universal solution to Transition Problem 1.

2 Preregistration Boundary and Pilot Disclosure

No B2 confirmatory seed had been executed, inspected or analysed when this record was frozen. Twenty held-out confirmatory seeds were generated independently of the development pilot and

were registered before confirmatory execution.

Three development-only seeds, `311221001`, `311221002` and `311221003`, were used to validate the implementation and select pilot-informed disturbance regimes, causal-edge threshold, rescue criteria and bypass margins. Those pilot outcomes are excluded from confirmatory evidence.

The development pilot showed that passive persistence could survive low disturbance, became unreliable under medium disturbance and failed under high disturbance; internal architectures survived withdrawal of external support; C1 ablation in the reciprocal architecture produced a strong C2 deficit; specific replacement of the nominated C1-to-C2 function restored C2 whereas an equal-budget rescue directed to C3 did not; and a budget-matched brute-force controller was a serious bypass capable of matching or exceeding reciprocal maintenance on ordinary viability. These observations informed the frozen protocol but cannot be counted as confirmatory evidence.

Confirmatory execution is prohibited until the final preregistration has been made public. After public registration, only the external execution-registry gate may change to `CLEARED_FOR_CONFIRMATORY_EXECUTION`. The hypotheses, manifests, seed set, thresholds, budgets, interventions and decision rules remain frozen.

3 Artificial Maintenance System

The system contains three essential functional components, C_1, C_2 and C_3, with bounded states

$$q_i(t) \in [0, 1]. \tag{2}$$

All components begin at state 1.0. The preregistered viability threshold is

$$\theta_i = 0.35 \tag{3}$$

for every component. The operational viability region is

$$\mathcal{V} = \{q : q_i \geq 0.35 \text{ for all essential } i\}. \tag{4}$$

The discrete-time update has the form

$$q_i(t+1) = \mathrm{clip}_{[0,1]}\left[q_i(t) - d_i - \xi_i(t) + m_i^{\mathrm{int}}(t) + m_i^{\mathrm{ext}}(t)\right], \tag{5}$$

where d_i is uncontrolled degradation, $\xi_i(t)$ is the registered stochastic disturbance, $m_i^{\mathrm{int}}(t)$ is internally generated maintenance and $m_i^{\mathrm{ext}}(t)$ is externally supplied maintenance.

Active architectures share total maintenance budget 0.03 per simulation step and target state 0.90 unless a registered intervention states otherwise.

4 Experimental Architectures

Five architecture classes are frozen.

1. **Architecture A: Passive persistence.** No internal or external maintenance is supplied. The maintenance budget is zero.
2. **Architecture B: External maintenance.** A deficit-based external controller supplies maintenance with total budget 0.03.
3. **Architecture C: Internal feed-forward maintenance.** Internal edges are $C_1 \to C_2$ and $C_1 \to C_3$. Forty percent of the total budget is reserved for source-node self-maintenance when required, with total maintenance limited to 0.03.
4. **Architecture D: Reciprocal internal maintenance.** The internal maintenance graph is strongly connected: $C_1 \to C_2 \to C_3 \to C_1$, with total maintenance budget 0.03.
5. **Architecture E: Budget-matched brute-force bypass.** Direct deficit-based internal repair uses total budget 0.03 without a reciprocal maintenance graph. This is a serious null/bypass rather than a weak baseline.

Architectural connectivity is not itself treated as evidence of causal maintenance. Candidate maintenance edges must survive intervention and rescue.

5 Disturbance Regimes

Table 1: Frozen disturbance regimes.

Regime	Horizon	Degradation	Noise max	Shock p	Shock magnitude
Low	400	0.0007	0.0002	0.010	0.02–0.05
Medium	400	0.0010	0.0003	0.015	0.03–0.06
High	400	0.0015	0.0005	0.040	0.04–0.09
Intervention	600	0.0010	0.0003	0.015	0.03–0.06

Matched seed identities generate matched stochastic disturbance sequences across architectures whenever they share the same environment definition.

6 Registered Confirmatory Seeds

Twenty held-out seeds were generated using NumPy `default_rng`/PCG64 initialised with generator seed `20261001`. They are disjoint from the three pilot seeds and are reused across matched conditions to support paired comparisons.

254443587	676812453
494519943	298946929
141790924	462690370
672220313	386034419
473136542	589534568
366847981	730433096
172571491	884327383
813038724	886430260
193027560	866196950
502749228	243163872

7 Registered Interventions

Four intervention structures are frozen.

1. **External-support withdrawal.** External maintenance is set to zero from simulation step 200 onward.
2. **C1 ablation.** C_1 is fixed at zero from step 200 onward; its own maintenance and outgoing source contribution are disabled according to the frozen implementation.
3. **Specific rescue.** Following C1 ablation, external replacement of 0.012 per step is supplied to C_2 from step 200 onward, representing the nominated $C_1 \to C_2$ maintenance function.
4. **Irrelevant rescue.** The same replacement budget, 0.012 per step, is instead supplied to C_3.

The preregistered causal-edge threshold is

$$\epsilon_w = 0.25. \tag{6}$$

8 Primary Quantities

No aggregate organisation, closure or constraint score is constructed. The registered outcomes are kept distinct:

- path viability through the registered horizon;
- terminal viability;
- terminal minimum component state;
- time to first failure, right-censored at the registered horizon;
- component-specific terminal states;

- paired ablation deficits relative to intact matched trajectories;
- topology-selectivity difference between C2 and C3 ablation deficits;
- specific and irrelevant rescue effects;
- maintenance expenditure as a budget-control diagnostic.

Repeated simulation steps are nested observations. The independent confirmatory replication unit is the registered seed within an architecture/condition.

9 Primary Hypotheses and Decision Rules

9.1 H1.1: Active Maintenance versus Passive Persistence

Claim. Active maintenance provides a selective advantage over passive stability when disturbances require continuing correction.

Retain if all of the following hold:

- passive path viability under low disturbance is at least 16/20;
- medium passive path viability lies between low and high passive viability;
- passive path viability under high disturbance is at most 4/20;
- at least three of the four active architectures are path viable in at least 16/20 high-disturbance seeds;
- mean active-minus-passive high-disturbance path-viability advantage is at least 0.60.

Restrict if active advantage is reproducible but the graded-disturbance or architecture-breadth criteria are not all met. **Suspend** if passive low-disturbance path viability is below 16/20 because the passive comparator would be too fragile for the intended graded-demand test. **Reject** if passive persistence matches active systems under high disturbance after matched controls.

9.2 H1.2: Internal versus External Maintenance

Claim. Internally maintained systems retain greater operational viability than externally maintained systems after external support is withdrawn.

Retain if external terminal viability is at most 4/20; feed-forward, reciprocal and brute-force internal architectures are each terminally viable in at least 16/20 seeds; and at least two internal architectures exceed the external mean terminal minimum-state by at least 0.25.

Restrict if an internal advantage is reproducible but one or more magnitude or consistency thresholds are missed. **Reduce/Reject** if support withdrawal does not selectively disadvantage external maintenance or external and internal systems show equivalent autonomous persistence.

9.3 H1.3: Reciprocal Topology and Targeted Failure Propagation

The primary reciprocal $C_1 \to C_2$ effect is the paired intact-minus-ablated terminal C2 state.

Retain if:

- mean paired reciprocal $C_1 \to C_2$ effect exceeds $\epsilon_w = 0.25$;
- at least 16/20 seeds have C2 deficit at least 0.30;
- in at least 16/20 seeds, reciprocal C2 deficit exceeds C3 deficit by at least 0.20;
- feed-forward ablation shows a different signature, with absolute C2-versus-C3 deficit difference at most 0.15 in at least 16/20 seeds.

Restrict if a reciprocal downstream effect is reproducible but topology selectivity misses one or more thresholds. **Reduce** if feed-forward or brute-force alternatives reproduce the same propagation signature. **Reject** if the reciprocal edge effect does not exceed ϵ_w or the predicted propagation repeatedly fails.

9.4 H1.4: Specific Functional Rescue

Claim. Specific functional replacement rescues the predicted downstream deficit caused by removal of C1.

Retain if:

- specific rescue raises terminal C2 by at least 0.30 on average relative to no-rescue C1 ablation and by at least 0.30 in at least 16/20 seeds;
- specific rescue exceeds equal-budget irrelevant rescue in terminal C2 by at least 0.30 on average and in at least 16/20 seeds;
- irrelevant rescue improves C2 by less than 0.10 on average relative to no-rescue ablation.

Restrict if specific rescue is reproducibly beneficial but magnitude/selectivity thresholds are not all met. **Reject** if matched replacement fails systematically or irrelevant rescue performs equivalently.

9.5 H1.5: Boundary and Timescale Robustness

Deferred and not tested in this first B2 preregistration. The present experiment fixes one three-component system boundary and one native simulation resolution. No claim of boundary or timescale robustness will be inferred from this study. A later preregistered robustness experiment is required before promoting such a claim.

9.6 H1.6: Explanatory Gain beyond Maintenance Budget / Serious Bypass

Architecture E is preregistered as a serious bypass lacking reciprocal topology.

Reduce the necessity of reciprocal topology if all noninferiority conditions hold:

- absolute high-disturbance path-viability difference between brute-force and reciprocal architectures is no greater than 0.10;
- absolute mean terminal minimum-state difference is no greater than 0.10;
- after C1 ablation, brute-force mean state across non-ablated C2/C3 is no more than 0.05 below reciprocal.

A **Reduce** verdict on reciprocal necessity does not negate a **Retain** verdict for a causal reciprocal edge under H1.3 or H1.4.

Retain explanatory gain within the tested architecture class only if reciprocal maintenance exceeds the bypass outside those margins and H1.3/H1.4 also retain. **Restrict** if any reciprocal advantage occurs only in the registered perturbation context. **Reject** if serious matched alternatives reproduce all decisive outcomes and no selective causal contribution of the proposed internal organisation survives.

10 Statistical and Decision Analysis

Primary inference is based on the frozen thresholds above, not on post-hoc null-hypothesis significance testing. Wilson 95 percent intervals are reported for viability proportions. Continuous paired effects use paired percentile-bootstrap 95 percent intervals with 10,000 resamples and fixed analysis RNG seed `20261002`. Failure-time summaries are right-censored at the registered horizon and reported descriptively.

Only C1 is preregistered for decisive node-level ablation; consequently no primary familywise multiple-comparison correction is required for the edge test. Any later C2 or C3 ablation analysis is exploratory unless separately preregistered.

The study explicitly permits mixed architectural outcomes. Internal maintenance may be retained as a sufficient artificial causal route even if reciprocal topology receives a Reduce verdict as a necessity claim. Conversely, a real reciprocal edge and successful specific rescue do not by themselves establish that reciprocal closure is required for maintenance.

11 Data Inclusion, Invalid Runs and Missing Data

All valid confirmatory runs from all 20 preregistered seeds are included. No run is excluded because it is surprising, unfavourable or an outlier in scientific outcome.

A run is classified as `INVALID_RUN` only for preregistered technical reasons such as manifest

or configuration hash mismatch, unregistered seed, environment-validation failure, non-finite numerical state, corrupted required output or failed provenance/integrity validation. Invalid runs remain in the audit trail and may be rerun with the same registered seed after the technical cause is corrected.

No statistical imputation is planned. Missing decisive evidence caused by unresolved technical failure leads to a Suspend judgement where appropriate rather than silent deletion or replacement with a new seed.

A scientifically negative valid run remains evidence:

$$\boxed{\texttt{VALID_NEGATIVE} \neq \texttt{INVALID_RUN.}} \tag{7}$$

12 Confirmatory Manifest Integrity

Seven confirmatory manifests are frozen under the source baseline identifier `STAGE_B2_0_7_0_PREREG_CANDIDATE`. Their manifest hashes are listed below.

Experiment	Version	Manifest SHA-256
B2_ABLATE_C1_CONFIRM_001	1.0.0	001debd3d4113afe399a6cb4ca36576e450ac2d5585f1ef60bd6726e646c4ee0
B2_MAINT_HIGH_CONFIRM_001	1.0.0	cdf42bc89d6636e92cae228deb5976654917ec2a3bc55d2e3b6a3748d5af3708
B2_MAINT_LOW_CONFIRM_001	1.0.0	96ccd4d8c304889527b815af745b03d8ce3004fb577cbc2f4542fb4b8476df37
B2_MAINT_MEDIUM_CONFIRM_001	1.0.0	bde254b1f35a709af1b5f4afd411f4b37f2210a767319aaa91b78ec1af0edecb
B2_RESCUE_IRRELEVANT_CONFIRM_001	1.0.0	74cf3761b00292362c9ea6079769f8ba6c54d56e94739dee715d02ef88d9a8c4
B2_RESCUE_SPECIFIC_CONFIRM_001	1.0.0	dfcb631dff80ca797f0d5b82efb0ce38d48904cee9500452e475791dce201607
B2_SUPPORT_WITHDRAW_CONFIRM_001	1.0.0	d0700f4472ca6cec075d37ff50611d954dcc5706754e684021c62e79dd7e0b7c

The lock files additionally contain SHA-256 hashes for referenced architectures, environments and registries. Any post-freeze alteration invalidates confirmatory execution.

13 Relationship to B1 and Wider Programme

B1 has already established, under preregistration, that sustained non-equilibrium Gray–Scott dynamics can support reproducible persistent spatial organisation. B2 does not treat that result as self-maintenance. Instead, it asks the distinct causal question of whether internally generated maintenance contributes selectively to preserving operational organisation.

The preceding B1 records are:

- OSF preregistration: 10.17605/OSF.IO/G3TE5;
- software/reproducibility package: 10.5281/zenodo.23075692;
- confirmatory dataset/decision record: 10.5281/zenodo.23075718.

B2 remains a test of Transition Problem 1, not an attempted recreation of the origin of life. Even a strong positive result would establish only that an artificial system can causally contribute to maintaining the organisation required for its continued operation within the tested conditions.

14 Post-Registration Execution Rule

Before public OSF registration, all seven experiment-registry entries remain `FROZEN_PREREGISTRATION_READY_UNRUN`. After the registration becomes public, its DOI and URL will be recorded as the authority for changing only those external gate entries to `CLEARED_FOR_CONFIRMATORY_EXECUTION`.

No confirmatory scientific field may be changed during that unlock. If a substantive change becomes necessary, a new experimental version and, where required, a new preregistration must be created.